

An Overview of Current Trends in Swarm Robotics

Ulrich Christian Huber

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Abstract

As decentralisation is asserting itself over monolithic design in wide-scale applications of robotics with ever smaller, yet more computationally powerful units, coordinating such robotic swarms effectively still poses a considerable challenge. Drawing from recent research, this paper aims to give an overview over the field of swarm robotics in general and the considerations of designing such systems in particular.

1 Introduction

With the advent of ever more powerful, yet smaller devices, computation in a distributed fashion is becoming more and more attainable in the realm of robotics. Leveraging this miniaturisation to develop more independent and replaceable systems with fewer significant points of failure beckons researchers to tackle hitherto unthinkable challenges in robotics to allow through smart machines working cooperatively with relatively little human input, freeing up humans from tasks such as monitoring or even physical labour. Swarm robotic approaches are in a uniquely advantageous position to make use of these advancements that, with the proper techniques and hardware requirements being met, could enable emergent behaviours to help with tasks previously beyond the reach of traditional architectures. However, building and maintaining these systems, moving from simulation to the real world, communications and other physical, digital considerations pose significant roadblocks. In order to give an overview of what is possible, what is yet to be solved and the possibilities on the horizon, current research topics and the state of the art are collected in this paper, attempting to elucidate the following questions:

- **RQ1:** *Which trends and approaches in the field of swarm robotics are currently being investigated?*
- **RQ2:** *Which gaps are present?*
- **RQ3:** *Has swarm robotics evolved beyond simple organisation towards complex behaviours?*

In order to answer these questions in a way that outlines the current research landscape, papers from 2018 and onwards have been prioritised and collected through search engines (Google Scholar, DBPL), though review papers referenced include older publications.

2 Robot Swarms

In a 2019 paper [4], Arnold et al. attempted to provide a general definition of swarm robotics to build upon. Generally, swarms are defined by several key characteristics: A collective, not necessarily homogenous grouping of robots composed of multiple individual units, robot swarms operate with only limited human intervention towards achieving a common goal or task. Human input is often limited to setting the goal rather than control over Communication can be explicit via dedicated communications channels or implicit via environmental information of manipulation and tasks are achieved through emergent behaviour.

2.1 Swarm Types

2.2 Swarm Behaviours

Tasks in the context of swarm robotics can take many forms. [15] provides a taxonomy of common task types and their properties. The most basic tasks fall within the category of spatial organisation, making swarm members aggregate in an area, form patterns, establish structures through self-assembly or clustering around object for further manipulation.

Building on top of movement, navigation tasks offer a more involved type of movement, whereby collective tasks such as self-organised exploration, formation movement, cooperative transport of objects or localisation of objects can be achieved, requiring a measure of coordination.

Decision making behaviours require communication capabilities, whether explicit or implicit. Distributed consensus mechanism, task allocation, collective fault detection and perception, synchronisation and group size regulation fall within this category.

Further general task types can be found in self-healing (recovering from faults), self-reproduction (creation of additional units) and human-swarm interaction.

3 Swarm Coordination

One of the primary drivers behind robotic swarm technology lies in lowering human intervention by implementing a higher degree of autonomy within individual units. Central to this task is the ability to self-organise what each robot should be doing at any given point in time. To this end, different approaches can be identified in literature, as outlined in this section.

3.1 Competitive Methods

Inspired by particle swarm optimisation but incorporating a pairwise competition mechanism, competitive swarm optimisation (CSO) [8] provides a linear time solution to optimisation problems in the realm of high-dimensional task allocation. In [7], Chauhan et al. investigate a ten year span of CSO research

3.2 Cooperative Methods

Particle Swarm Optimisation

Inspired by bird-like flocking behaviour, particle swarm optimisation (PSO) [12] is a metaheuristic approach that moves candidate solutions as particles in a search space to improve a given quality metric. In the realm of swarm robotics, PSO has enjoyed increasing interest, especially regarding the target search problem, as identified by Hamami and Ismail [10]. They noted a significant up-tick of publications in research interest of PSO in swarm robotics compared to other fields, noting a frequent approach of mapping individual swarm members onto particles. Adaptations of PSO have found their way into land, air and sea based swarm robot control mechanisms and path planning. However, open challenges are still present in sweet spot balance (diversification vs intensification), lack of unified mathematical model structure for analysing PSO performance metrics and lack of testing environments that closely mimic real world constraints.

Ant Colony Optimisation

Bacterial Foraging Optimisation

Artificial Bee Colony

Artificial Fish Swarm

[14] [17]

4 Computer Assisted Swarm Design Methodologies

In general, evolutionary design can be classified along two axes: Automatic approaches, in which an optimisation process generates control software without human input beyond the initial formalisation of the underlying model, differ from semi automatic approaches, where humans remain in the loop and guide the optimisation process [13].

The other axis concerns itself with deployment. Offline methods generate control software before deployment, while online methods target an already ongoing robot configuration that is activated.

4.1 Automatic Design - AutoMoDe

In 2014, Francesca et al. investigated methods for automatic swarm robot control software generation that could close the reality gap, i. e. the difference

between simulated performance and physical, real world deployment. Using a combination of probabilistic finite state machines (PFSM) and evolutionary F-Race optimisation, their approach proved to fulfill this goal remarkably well, especially compared to other evolutionary approaches as outlined in an empirical study conducted by Hasselmann et al. [11]. AutoMoDe variations are often being referred to as flavours partly due to its original naming convention of "Vanilla" and "Chocolate", with several descendant variants employing similar nomenclature. Table gives an overview of select variants featuring iterations on the original concept.

Year	AutoMoDe Flavour	Target Robot Platform	Control Model Type	Communications Module
2014	Vanilla	E-puck	PFSM	None
2015	Chocolate	E-puck	PFSM	None
2018	Maple	E-puck	Behaviour Trees	None
2018	Gianduja	E-puck	PFSM	Signifier Broadcast
2019	Waffle	E-puck	PFSM	Infrared Transceiver
2019	Icepop	E-puck	PFSM	None
2020	Gianduja2/Gianduja3	E-puck	PFSM	Signifier Broadcast
2020	TuttiFrutti	E-puck	PFSM	Colour Detection/E
2020	Arlequin	E-puck	PFSM	None
2021	Cedrata	E-puck	Behaviour Trees	Range-and-bearing
2022	Cedrata-GP/Cedrata-GE	E-puck	Behaviour Trees	Range-and-bearing
2023	Nata	E-puck	PFSM	None
2024	Habanero	E-puck	PFSM	Artificial pheromone

It is of particular note that so far, only E-Puck has been used as a platform, ports to other robots could not be found, leaving open the challenge of a platform-agnostic implementation. However, this should be

4.2 Semiautomatic Design - Human in the loop

In contrast to fully automatic design methods for general use purposes, more complex and specific use cases can benefit from manual guidance in the process of generating control software. [5]

5 Application Domains and their challenges

As robots are deployed in diverse environments, they face domain-specific problems that impose restrictions on their use and require task- and environment-specific considerations before deployment.

5.1 Unmanned Aerial Vehicles

Operating under 6 Degrees of freedom (DOF) as they are not constrained by terrestrial movement, swarms of robotic unmanned aerial vehicles (UAV) face

additional challenges, especially in terms of orchestrating movement. A recent review by Alqudsi et al. [3] mentions aerial collisions as a particular challenge for robots moving in formation in a shared environment, requiring orchestration of movement in centralised or decoupled planning modes. Centralised planning represents a group of robots as one unit, whereby the DOF of individual agents are considered together and the configuration space must be searched for valid path combinations. This, however, comes at the cost of linear complexity, limiting scope.

Decoupled planning, on the other hand, features a planner generating routes for each agent by itself, resulting in a collision-free path for every robot. Speed is then adjusted per unit. The lower dimensional search space enables a more scalable solution. Further challenges arise in the form of vulnerabilities in terms of secure communications and hostile actions. Due to signal requirements for coordination, they are susceptible to jamming, spoofing and eavesdropping. Secure channels and countermeasures to interference are considerations when designing such systems.

Another review of aerial swarm robotics can be found in [1]

5.2 Swarm Intelligence and Robotics in Agriculture

In the sector of agriculture, [16] conducted a review of relevant literature have identified several developments concerning swarm intelligence - quite literally - in the field. While the study is mostly focused on, a section on swarm intelligence and swarm robotics singles out particular [6] in particular for its use of ground and aerial UAVs traversing agricultural topology and the challenges associated with it.

Further research indication is given in the direction of autonomous path navigation and particular attention to the remote nature of farming posing practical challenges regarding connection as well as security challenges concerning unauthorised access to the robots, digitally and physically. Another review by Albiero et al. [2] indicates a lack of scientific interest for agriculture as a focus area and a low technological readiness level for swarm robotics with less than a fourth of surveyed studies showing prototypes being put to use in operational environments, implying significant potential for impactful research. Conditions harsh towards delicate robotic parts, hostile to wireless communications, unstructured environments with crops as soft, easily damaged and malleable targets are identified as challenges particular to mechanised agriculture, requiring improvements in hardware, complex collective emergent behaviours to compensate. They also recommend further investigation of swarm strategies tackling complex topologies of farmland contour, shape and crop configurations. /new-line Under the umbrella of adaptability, handling uncertainty through enhanced self and exterior perception combined with improved guidance systems are raised as issues pertaining to highly variable environments: Lighting, irregular reliefs and discontinuous geometry. In the same vein, tackling distributed computing is encouraged as smooth communications cannot be guaranteed.

Lastly, autonomy is raised as an area of interest: Robots working in unstruc-

tured environments will require control architectures tailored to uncertainty. To that end, reinforcement learning is singled out as a strong candidate for further research into sub-dividing behaviours and modularising the learning process. The general consensus seems to be that agriculture as a whole is underserved in the research community and that it would benefit greatly from prototyping in more specific environments closer to reality due to the harder challenges posed by them.

5.3 Underwater Swarm Robotics

[9]

6 Conclusion

Swarm robotics as a field is evolving rapidly, yet industrial application and. While centralised approaches featuring swarm intelligence components do appear, the core idea of distributed, autonomous swarm behaviours have yet to find widespread adoption [17].

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